

Permutations

Date _____ Period _____

List all possible permutations.

1) 1, 2, 3

2) T, V, W, taken two at a time

3) ☺, ☀, ♥, ▲, taken two at a time

4) A, B, C, D

Evaluate each expression.5) ${}_4P_2$ 6) ${}_5P_1$ 7) ${}_6P_2$ 8) ${}_6P_6$ 9) $-4 + {}_7P_4$ 10) $5 \cdot {}_6P_5$ **Find the number of unique permutations of the letters in each word.**

11) DESIGN

12) MATH

13) CHEESE

14) FURTHER

15) BALLISTICS

16) BILLIONAIRE

Critical thinking questions:

17) Write a word for which there are 30 unique permutations of the letters.

18) Simplify ${}_xP_2$